

11/22/22 Meeting Notes

11/30/22 Updated with additional information provided by Erik Olson via email.

Meeting with City of Cadillac staff Jeff Dietlin (DPW Director) and Patrick Miller (WWTP Operator), and Erik Olson of Cadillac Casting Inc.

The meeting was intended to discuss the processes at the Cadillac Casting Inc (CCI) facility. Previous permits issued to CCI by the City of Cadillac have classified the facility as a Part 464, Subpart C CIU, however the permit does not include any categorical limits or other appropriate elements for a CIU with production-based limits.

CCI is a ductile iron casting foundry only. Erik Olson stated that the volume of metal cast significantly exceeds 3,557 tons a year.

Erik Olson had reviewed the categorical part and identified the Furnace scrubber standard as applicable with 3 in series scrubber operations which discharge to the sanitary sewer via treatment process (a primary quench scrubber, venturi scrubber, and demister). There is also a slag quench process for the cupola which discharges to the same treatment system. In follow up correspondence on 11/30/22 Erik clarified that the slag quench process does not discharge to treatment. City water is only used to replenish the water lost to evaporation through this process. The cupola and the holding furnaces are cooled with NCCW which discharges to the sanitary sewer and combines with the categorical waste prior to discharge.

In addition, the facility has 2 stand alone (not in series) wet dust scrubbers which discharge to the sanitary sewer via a second treatment process.

Both treatment systems include a sludge press, the wastewater from which is discharged and combines with NCCW and sanitary waste before discharge to the sanitary sewer. There is currently only one sampling point after all flows combine. The facility and the city would like to maintain a single sampling point, if possible. More information on the flow volumes and pollutant concentrations from other processes would be required to determine adjusted limits at this sample point sampling point.

Sand is used in the facility for molds (green sand) but is not reclaimed.

At this time there are no sample points at the end of process or flow monitoring for each process/treatment system, although the facility is willing to add this, if necessary, but will need to hire a consultant to assist them with appropriate flow metering technology. An improved flow diagram has been developed for the CCI facility – I requested a copy of this diagram to better understand where flows combine and where sampling points are proposed.

The facility was originally constructed in 1922, however the scrubber processes were installed in the mid-1990's. The PSNS limits appear to be applicable to these processes.

The facility does have production data available for both the mass of metal poured and the volume of air scrubbed. It is likely that the mass limits will be easily determined, however flow data is required to determine compliance at the sample point and would also be required to determine adjustment of the limit if sampled after categorical flow combines with the NCCW and sanitary flow.

The City of Cadillac is currently conducting a MAHL study and is reviewing data to determine if any changes are appropriate to the local limits. After local limits have been adjusted as necessary the permits will be reviewed.